Sujal Roy

Mechatronics Engineer

RESUME_1511

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Total Experience: 13 years

PROFESSIONAL SUMMARY

Mechatronics Engineer with 13 years of experience in product development, using CAD, SolidWorks, Arduino, ESP32. Experienced in electromechanical systems, integration tests and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Testing, ESP32, Motor Drivers, SolidWorks, Python, Sensors, Arduino, PLC, CAD, MATLAB, C++, Control Systems

Methods: Electromechanical systems, Integration tests, Product prototyping

PROFESSIONAL EXPERIENCE

Lead Mechatronics Engineer | Ironleaf Engineering | Jan 2022 - Present

- Improved prototype throughput by 39% through actuator timing optimization
- Reduced assembly cost by 18% after redesigning electromechanical fixtures
- Validated 695 integration test cases across mechanical and electrical interfaces
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Mechatronics Engineer | Asterline Engineering | Jan 2017 - Dec 2021

- Validated 651 integration test cases across mechanical and electrical interfaces
- Raised measured actuator positioning consistency by 35% after calibration

Associate Mechatronics Engineer | Urbanaut Engineering | Jul 2013 - Dec 2016

- Improved prototype throughput by 19% through actuator timing optimization
- Reduced assembly cost by 27% after redesigning electromechanical fixtures

PROJECTS

Smart Dispensing Unit | Control Systems, C++, Motor Drivers, Testing

- Built controller firmware, mechanical enclosure and fault signals.
- Documented acceptance criteria and evaluated the design using integration tests and product prototyping.

Automated Sorting Prototype | Testing, Control Systems, C++, Arduino

- Integrated optical sensors, actuator timing and conveyor control.
- Documented acceptance criteria and evaluated the design using integration tests and product prototyping.

EDUCATION

B.Tech in Mechatronics Engineering, Western Plains Institute of Technology, 2013
Relevant Methods: Electromechanical systems, Integration tests, Product prototyping

CERTIFICATIONS

Rapid Prototyping Workshop (2025)